YELLOW REPORT

Testosterone replacement therapy (TRT) can be considered in this patient because he has **favorable pathology** (negative margins, no nodal disease, Gleason 4+3) and an **undetectable PSA for 12 months** after radical prostatectomy, and there is **no evidence of increased biochemical recurrence** with TRT in this setting [1] [2]. However, this is a **shared decision** with explicit counseling that the evidence is limited and the oncologic risk is unquantified [1]. If TRT is initiated, use a **structured monitoring plan**: check PSA and hematocrit at 3, 6, and 12 months, then annually, and stop TRT if PSA rises or other concerns arise [3] [4]. Symptomatic benefits are likely within 3–6 months if TRT is appropriate for his symptoms [5].

Evidence on oncologic safety of TRT after radical prostatectomy

- **Favorable pathology**: Negative margins, no nodal disease, and Gleason 4+3 (Grade Group 3) are associated with lower recurrence risk after radical prostatectomy.
- **Undetectable PSA**: Sustained undetectable PSA for 12 months indicates no biochemical recurrence, supporting a lower risk profile.
- **TRT studies**: Multiple studies show no increased biochemical recurrence with TRT in men with favorable pathology and undetectable PSA after radical prostatectomy [1] [2].
- **Guideline stance**: The AUA states TRT can be considered in men with favorable pathology and undetectable PSA after radical prostatectomy, with no demonstrated increase in biochemical recurrence [1].

Expected symptomatic benefits of TRT

TRT can improve symptoms of hypogonadism, including fatigue, low libido, and mood disturbances. Benefits typically appear within **3–6 months** of therapy, with improvements in energy, sexual function, and overall quality of life [5]. Bone density, muscle mass, and strength may also improve with longer-term therapy.

Monitoring plan if TRT is initiated

If TRT is initiated, a **structured monitoring plan** is essential:

Parameter	Monitoring frequency
PSA	At 3, 6, and 12 months, then annually

Parameter	Monitoring frequency
Hematocrit	At 3, 6, and 12 months, then annually
Testosterone levels	Every 6 months initially, then annually
Clinical symptoms	At each visit

Stop TRT if PSA rises or other concerns arise, and reassess the risk-benefit balance.

Alternative strategies if TRT is declined

If TRT is declined, consider:

- **Lifestyle interventions**: Exercise, weight loss, and sleep optimization may partially alleviate symptoms.
- **Non-testosterone therapies**: Address specific symptoms (e.g. PDE5 inhibitors for erectile dysfunction, antidepressants for mood).
- **Continued surveillance**: Monitor symptoms and testosterone levels, and reassess if symptoms worsen or new evidence emerges.

Shared decision-making and patient counseling

Emphasize that the decision is **shared** and individualized, with explicit counseling that the oncologic risk is unquantified and the evidence base is limited [1]. Discuss the potential for symptom improvement, the need for close monitoring, and the possibility of discontinuing TRT if PSA rises or other concerns arise.

Sources:

1. Evaluation and management of testosterone deficiency: AUA guideline. The Journal of Urology, 2018
2. Testosterone replacement therapy following definitive treatment for prostate cancer: a scoping review of safety and efficacy. International Journal of Impotence Research, 2025
3. Prostate safety events during testosterone replacement therapy in men with hypogonadism: a randomized clinical trial. JAMA Network Open, 2023
4. Testosterone therapy in men with hypogonadism: an endocrine society clinical practice guideline. The Journal of Clinical Endocrinology and Metabolism, 2018
5. Approach to the patient: the evaluation and management of men ≥ 50 years with low serum testosterone concentration. The Journal of Clinical Endocrinology and Metabolism, 2023